Concept 3 | Applications of Differentiation: Polynomial Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Rate

$$\frac{dy}{dx}$$

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

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Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q001 Written

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Solution

1. Differentiate, solve $f'(x) = 0$, make a sign table, and substitute each stationary x-value into $f(x)$.

Final answer

Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.

Q002 Written

Find the extreme values and sketch the three source Try cubics.

Solution

1. Record the sign of f' from left to right and label each change $+ \to -$ as a local maximum and $- \to +$ as a local minimum.

Final answer

Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q003 Written

Find the extreme values and draw the graphs for all nine source cubic functions in Exercise.

Solution

1. Use exact stationary coordinates and keep the graph's increasing/decreasing intervals consistent with the sign table.

Final answer

Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.

Q004 Written

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

Solution

1. $P'(x) = 3[(x - a)^2 - 1]$, so the critical points are $a - 1$ and $a + 1$. The derivative changes $+ \to -$ then $- \to +$.

Final answer

Local maximum at $x = a - 1$; local minimum at $x = a + 1$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q005 Written

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Solution

1. Compare $f(-3), f(4)$ with the stationary values $f(-2) = -11$ and $f(2) = 21$.

Final answer

Maximum 21 at $x = 2$; minimum -11 at $x = -2$.

Q006 Written

Find the maximum and minimum values of the two source Try cubics on their stated closed intervals.

Solution

1. A local extreme is not automatically absolute: compare it with both domain endpoints.

Final answer

Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q007 Written

Find the maximum and minimum values of all six source cubics on their given closed intervals.

Solution

1. State the extreme value together with its x-coordinate and retain only points in the specified domain.

Final answer

For each function, compare the endpoint values with the values at all critical x-values lying in the interval.

Q008 Written

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

Solution

1. Critical points are $x = \pm\sqrt{2}$, with values $\mp 4\sqrt{2}$; compare these with $f(-3) = -9$ and $f(3) = 9$.

Final answer

Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q009 Written

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a, b, and the local minimum.

Solution

1. Use $f'(2) = 0$ to get $a = 12$, then $f(2) = 9$ to get $b = -7$. The other stationary point is $x = -2$.

Final answer

$a = 12, b = -7$; local minimum -23 at $x = -2$.

Q010 Written

Solve the two source Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

Solution

1. Differentiate first, use each stated extreme point, then evaluate the remaining extreme value from the resulting cubic.

Final answer

Use $f'(a) = 0, f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q011 Written

Solve all six source Exercise problems asking for cubic coefficients from extrema.

Solution

1. Do not skip the condition $f'(a) = 0$: it identifies the stationary point before the function-value equation is used.

Final answer

Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.

Q012 Written

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c.

Solution

1. The derivative roots are $-2, 4$, so $f'(x) = 3(x+2)(x-4) = 3x^2 - 6x - 24$. Thus $a = -3, b = -24$. Substitution of $f(-2) = 178$ gives $c = 150$.

Final answer

$a = -3,\ b = -24,\ c = 150.$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q013 **Written**

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k.

Solution

1. The derivative is $f'(x) = 3x^2 - 6kx + k$. No two distinct stationary roots means $D = 36k^2 - 12k \leq 0$, hence $0 \leq k \leq \frac{1}{3}$.

Final answer

$0 \leq k \leq \frac{1}{3}$.

Q014 **Written**

Determine the parameter ranges in the source Try problems for a cubic to have no extreme values or to have both extremes.

Solution

1. Differentiate, calculate the quadratic discriminant, and solve the resulting parameter inequality exactly.

Final answer

Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q015 Written

Determine the parameter ranges for all six source Exercise cubics according to whether they have extreme values.

Solution

1. Remember: $D > 0$ gives two distinct real stationary roots; $D \leq 0$ gives no pair of distinct extremes.

Final answer

Apply the derivative-discriminant criterion separately to each printed cubic.

Q016 Written

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k-range that guarantees both a local maximum and minimum.

Solution

1. $f'(x) = 3x^2 + 6kx + 12k$ has two distinct real roots when $36k^2 - 144k > 0$, i.e. $k(k-4) > 0$.

Final answer

$k < 0ork > 4$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q017 Written

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k.

Solution

1. For a positive leading coefficient, monotonic increase requires the derivative quadratic to have no two distinct real roots: $D = 24k^2 - 60 \leq 0$.

Final answer

$-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$.

Q018 Written

Find the parameter ranges in the two source Try problems for the cubic to be monotonically increasing.

Solution

1. Monotonicity is checked from $f'(x) \geq 0$ for every real x; use the discriminant condition for the quadratic derivative.

Final answer

Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q019 **Written**

Find the parameter ranges in all six source Exercise problems for monotonic increase.

Solution

1. Complete the discriminant test and include equality when the derivative has a repeated zero.

Final answer

Differentiate each cubic and impose that its quadratic derivative never becomes negative.

Q020 **Written**

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k+6)x + 10$, find the k-range for monotonic increase on $x \in [1, \infty)$.

Solution

1. $f'(x) = x^2 - 2kx + k + 6$. If $k \leq 1$, its minimum on $[1, \infty)$ is $7 - k \geq 0$; if $k > 1$, the vertex condition gives $k + 6 - k^2 \geq 0$, hence $k \leq 3$.

Final answer

$k \leq 3$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q021 **Written**

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.

Solution

1. The stationary values are $f(1) = 1 > 0$ and $f(3) = -3 < 0$, so the cubic crosses the x-axis three times.

Final answer

Three distinct real roots.

Q022 **Written**

Find the number of distinct real roots for the three source Try equations.

Solution

1. Draw $y = f(x)$, find its critical points, and compare the local maximum/minimum with zero.

Final answer

Use the cubic graph: count x-axis intersections from the stationary points and end behavior.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q023 **Written**

Find the number of distinct real roots for all nine source cubic equations.

Solution

1. Differentiate, locate the turning points, and use their signs relative to zero to decide between one, two, or three distinct roots.

Final answer

Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.

Q024 **Written**

Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.

Solution

1. The derivative is $-3(h-2)(h-6)$; both stationary values are positive, so the graph crosses the axis only once.

Final answer

One distinct real root.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q025 Written

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a.

Solution

1. Rewrite as $x^3 - 3x^2 - 9x = a$. The local values are 5 and -27; a horizontal line intersects the cubic three times strictly between them.

Final answer

$-27 < a < 5$.

Q026 Written

Solve the two source Try parameter problems for three or two distinct real roots.

Solution

1. Strict inequalities give three roots; equality at a stationary value produces two distinct roots and must be handled separately.

Final answer

Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q027 Written

Solve all four source Exercise parameter problems for the required number of distinct roots.

Solution

1. Keep the endpoint cases: tangency at an extreme value counts as two distinct roots, not three.

Final answer

Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.

Q028 Written

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

Solution

1. Set $g(x) = x^3 - 6x^2 + 9x$, whose extrema are 0 and 4. The horizontal level is $2a^2 - 7a + 2$; require $0 < 2a^2 - 7a + 2 < 4$.

Final answer

$$\frac{7-\sqrt{65}}{4} < a < \frac{7-\sqrt{33}}{4} or \frac{7+\sqrt{33}}{4} < a < \frac{7+\sqrt{65}}{4}.$$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q029 Written

Prove the source inequalities using an auxiliary cubic and a derivative sign table.

Solution

1. Differentiate the auxiliary function, make the sign table on the domain, and substitute the minimum point or endpoint.

Final answer

Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.

Q030 Written

Prove all three source cubic inequalities on their stated domains.

Solution

1. The proof is complete only after the derivative sign table and the minimum value are stated.

Final answer

Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q031 Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

Solution

1. The maximum of $x^3 - 3x^2 - 3x + 3$ on the interval is $4\sqrt{2} - 2$ at $x = 1 - \sqrt{2}$. Hence $k + 4\sqrt{2} - 2 \leq 0$.

Final answer

$k \leq 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.

Q032 Written

Find the extreme values and sketch the two source quartic functions.

Solution

1. Differentiate the quartic, use a sign table for the cubic derivative, and substitute the critical values.

Final answer

For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q033 Written

Find the extreme values and graphs for all six source quartic functions.

Solution

1. Repeated derivative roots may be stationary without being extrema; check the sign change before labelling a point.

Final answer

Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.

Q034 Written

For $f(x) = x^4 - 4x^3 + k$, find the exact k-range for exactly two distinct real roots.

Solution

1. The global minimum of $x^4 - 4x^3 + k$ is $k - 27$ at $x = 3$. Exactly two crossings require this minimum to be negative.

Final answer

$k < 27$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q035 MCQ

At a local maximum, the sign of f' changes:

Solution

1. Use the first derivative sign change.

Correct option: B

Final answer

$+ to -$

Q036 MCQ

For $P'(x) = 3[(x-a)^2 - 1]$, the local maximum occurs at:

Solution

1. The derivative changes + to - at $a - 1$.

Correct option: A

Final answer

$a - 1$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q037 *MCQ*

Absolute extrema on a closed interval require comparing:

Solution

1. Closed intervals include both.

Correct option: C

Final answer

stationary points and endpoints

Q038 *MCQ*

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:

Solution

1. Compare the endpoints with the critical values.

Correct option: B

Final answer

9

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Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Solution

1. Set $f'(2) = -12 + a = 0$.

Correct option: C

Final answer

12

Q040 MCQ

The derivative roots in the challenge are:

Solution

1. Use the two turning-point x-values.

Correct option: B

Final answer

-2 and 4

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q041 MCQ

No two distinct extrema for a cubic means the derivative discriminant is:

Solution

1. A repeated root or no real roots gives no pair.

Correct option: C

Final answer

$D <= 0$

Q042 MCQ

The challenge range for both extrema is:

Solution

1. Require $36k^2 - 144k > 0$.

Correct option: B

Final answer

$k < 0 or k > 4$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Solution

1. Use $f'(x) \geq 0$.

Correct option: B

Final answer

non-negative everywhere

Q044 MCQ

The challenge domain in 7-13 is:

Solution

1. Use the restricted domain in the vertex test.

Correct option: B

Final answer

[1,infinity)

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q045 MCQ

Three distinct roots occur when the x-axis lies:

Solution

1. Then the graph crosses the axis three times.

Correct option: B

Final answer

between the extrema

Q046 MCQ

The 7-14 challenge has how many distinct real roots?

Solution

1. Both turning values are above zero.

Correct option: B

Final answer

1

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q047 MCQ

Three intersections with $y = a$ require the level to lie:

Solution

1. Tangency is an endpoint case.

Correct option: A

Final answer

strictly between the extrema

Q048 MCQ

The 7-15 warm-up range is:

Solution

1. Use the two local values.

Correct option: A

Final answer

$-27 < a < 5$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q049 MCQ

To prove LHS $\geq$ RHS, define:

Solution

1. Then prove $f \geq 0$.

Correct option: B

Final answer

LHS-RHS

Q050 MCQ

The largest k in the challenge is:

Solution

1. Bound k by the maximum of the k-free part.

Correct option: A

Final answer

2-4sqrt(2)

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q051 MCQ

A repeated root of a derivative may be:

Solution

1. Check the sign change.

Correct option: C

Final answer

stationary but not an extreme

Q052 MCQ

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:

Solution

1. The global minimum must be below zero.

Correct option: B

Final answer

$k < 27$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q053 MCQ

A local minimum is identified by the change:

Solution

1. A negative derivative followed by a positive derivative gives a local minimum.

Correct option: B

Final answer

$-to+$

Q054 MCQ

For $y = x^3 - 6x^2 + 16$, the critical x-values are:

Solution

1. Solve $3x^2 - 12x = 0$.

Correct option: A

Final answer

0 and 4

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q055 MCQ

The derivative sign table is used to determine:

Solution

1. Signs of (f') control monotonicity.

Correct option: B

Final answer

increase/decrease and extrema

Q056 MCQ

On a closed interval, an endpoint can be:

Solution

1. Compare both endpoints with stationary values.

Correct option: B

Final answer

an absolute extreme

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Solution

1. Solve $-3x^2 + 12 = 0$.

Correct option: A

Final answer

-2 and 2

Q058 MCQ

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Solution

1. Compare all candidates; $f(-3) = -9$.

Correct option: A

Final answer

$x = -3$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Solution

1. An extreme point is stationary.

Correct option: B

Final answer

$f'(a) = 0$

Q060 MCQ

In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:

Solution

1. The derivative changes from negative to positive at -2.

Correct option: A

Final answer

$x = -2$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q061 *MCQ*

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Solution

1. Use $D = 12k(3k - 1) \leq 0$.

Correct option: A

Final answer

$0 <= k <= 1/3$

Q062 *MCQ*

Two distinct stationary points occur when the derivative discriminant is:

Solution

1. A positive discriminant gives two distinct real derivative roots.

Correct option: C

Final answer

$D > 0$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Solution

1. The quadratic has no two distinct real zeros.

Correct option: A

Final answer

non-negative for all x

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:

Solution

1. Use the operational interval.

Correct option: A

Final answer

[1,infinity)

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q065 MCQ

A cubic with both stationary values above the x-axis has:

Solution

1. The graph crosses the x-axis once.

Correct option: B

Final answer

one distinct real root

Q066 MCQ

To count cubic roots, compare the extrema with:

Solution

1. Roots are x-axis intersections.

Correct option: B

Final answer

the x-axis

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:

Solution

1. One contact is tangent and the other crossing remains.

Correct option: B

Final answer

two distinct intersections

Q068 MCQ

For $x^4 - 4x^3 + k$, the global minimum occurs at:

Solution

1. The derivative is $4x^2(x - 3)$.

Correct option: C

Final answer

$x = 3$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Quiz WRITTEN 1 **Concept Quiz · Written**

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Solution

1. Use $f'(x) = 3x(x - 2)$ and a sign table.

Final answer

Maximum 1 at $x = 0$; minimum -3 at $x = 2$

Quiz WRITTEN 2 **Concept Quiz · Written**

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at x=1.

Solution

1. Use $f'(1) = 0$ and $f(1) = 4$.

Final answer

$a = 3,\ b = 2$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Quiz WRITTEN 3 **Concept Quiz · Written**

Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.

Solution

1. Set the derivative discriminant ≤ 0.

Final answer

$-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 **Concept Quiz · Written**

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

Solution

1. Use the maximum of the k-free cubic on the interval.

Final answer

$2 - 4\sqrt{2}$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Quiz MCQ 5 **Concept Quiz · MCQ**

At a local minimum, f' changes:

Solution

1. Use the first derivative sign change.

Correct option: B

Final answer

$-to+$

Quiz MCQ 6 **Concept Quiz · MCQ**

A cubic with a positive-leading derivative and $D < 0$ is:

Solution

1. The derivative quadratic stays positive.

Correct option: A

Final answer

always increasing

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Quiz MCQ 7 Concept Quiz · MCQ

The 7-15 challenge requires the level to satisfy:

Solution

1. Three distinct intersections lie strictly between extrema.

Correct option: B

Final answer

$0 < t < 4$

Quiz MCQ 8 Concept Quiz · MCQ

The global minimum of $x^4 - 4x^3 + k$ is:

Solution

1. Evaluate at the quartic's global minimum $x = 3$.

Correct option: A

Final answer

$k - 27$

Answer key

Use the question number shown on each page.

Q001 Use the derivative sign table: for x^3-6x^2+16, local maximum 16 at $x=0$ and local minimum -16 at $x=4$; classify the other two from their stationary points.

Q002 Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.

Q003 Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.

Q004 Local maximum at $x=a-1$; local minimum at $x=a+1$.

Q005 Maximum 21 at $x=2$; minimum -11 at $x=-2$.

Q006 Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.

Q007 For each function, compare the endpoint values with the values at all critical x-values lying in the interval.

Q008 Absolute minimum -9 at $x=-3$; absolute maximum 9 at $x=3$.

Q009 $a=12, b=-7$; local minimum -23 at $x=-2$.

Q010 Use $f'(a)=0, f(a)=M$, and the second stationary condition to form three simultaneous equations in the coefficients.

Q011 Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.

Q012 $a=-3,\ b=-24,\ c=150.$

Q013 $0\le k\le \frac{1}{3}.$

Q014 Use $D\le 0$ for no extremes and $D>0$ for two distinct stationary points.

Q015 Apply the derivative-discriminant criterion separately to each printed cubic.

Q016 $k<0 or k>4.$

Q017 $-\frac{\sqrt{10}}{2}\le k\le \frac{\sqrt{10}}{2}.$

Q018 Set the derivative discriminant $D\le 0$, with the derivative leading coefficient positive.

Q019 Differentiate each cubic and impose that its quadratic derivative never becomes negative.

Q020 $k\le 3.$

Q021 Three distinct real roots.

Q022 Use the cubic graph: count x-axis intersections from the stationary points and end behavior.

Q023 Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.

Q024 One distinct real root.

Q025 $-27<a<5.$

Q026 Rewrite each equation as $g(x)=a$, then compare the parameter line with the local maximum and minimum of g.

Q027 Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.

Q028 $\frac{7-\sqrt{65}}{4}<a<\frac{7-\sqrt{33}}{4} or \frac{7+\sqrt{33}}{4}<a<\frac{7+\sqrt{65}}{4}.$

Q029 Set $f(x)=\text{LHS}-\text{RHS}$, prove $f(x)\ge 0$ on the stated domain, and identify its minimum.

Q030 Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.

Q031 $k\le 2-4\sqrt{2}$; the largest value is $2-4\sqrt{2}$.

Q032 For $3x^4+4x^3-12x^2$: local maximum 0 at $x=0$, local minima -32 at $x=-2$ and -5 at $x=1$. For $-x^4-4x^3-8$: maximum 19 at $x=-3$; $x=0$ is stationary but not an extreme.

Q033 Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.

Q034 $k<27.$

Q035 B

Q036 A

Q037 C

Q038 B

Q039 C

Q040 B

Q041 C

Q042 B

Q043 B

Q044 B

Q045 B

Q046 B

Q047 A

Q048 A

Q049 B

Q050 A

Q051 C

Q052 B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at $x = 2$

Quiz WRITTEN 2 $a = 3,\ b = 2$

Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A